

## Vox Clamantis

# The Multitarget Drug Approach in Migraine Treatment: The New Challenge to Conquer

Abimael González-Hernández, PhD; Miguel Condés-Lara, PhD

Migraine is a complex neurovascular disorder where a complex and interrelated neuronal spinal, supraspinal and central mechanisms are involved. Although we have greatly advanced in the knowledge of the main pathways involved in this disorder, the current drugs used are not effective in all patients, suggesting that the key mechanism related to headache relief remains elusive. In this context, the multi-target drug approach or network pharmacology emerges as a new step in the development of innovative migraine pharmacotherapy. The design, discovery, and development of new drugs that reach several (instead of unique) specific targets (functional selectivity) involved in the migraine pathophysiology is essential to progress in the migraine treatment and open a new field of study about the main pathways and targets that could synergistically improve the migraine management.

**Key words:** network pharmacology, functional selectivity, pharmacotherapy

(*Headache* 2014;54:197-199)

In the last 25 years, the development of anti-migraine compounds following the rational drug design (ie, triptans and gepants) has allowed us to advance in the knowledge of specific pathways involved in the migraine pathophysiology.<sup>1-4</sup> Certainly, this approach has boosted the pharmacotherapy of several illnesses, making better the specificity of a compound for a specific receptor and avoiding undesirable effects and unspecific actions associated with inherent “promiscuity” of several drugs.

**From the Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Queretaro, Mexico.**

**Address all correspondence to A. González-Hernández, Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus UNAM-Juriquilla, Queretaro 76230, Mexico.**

Notwithstanding the fact that we have performed and developed a pharmacological arsenal to treat migraine headache, not all patients respond to a specific treatment,<sup>4,6</sup> suggesting at the first instance that the key mechanisms related to migraine relief remain elusive. This point appears obvious if we take literally the fact that migraine headache is the product of the interaction with complex and multifactorial variables,<sup>1-3,7</sup> and the current animal models used to understand its pathophysiology only reflect some characteristics of this disorder. Physiologically speaking, the pain control is a product of functional interactions between multiple anatomical structures, receptors, and neuromediators.<sup>8</sup>

Currently, migraine is considered as a debilitating and complex neurological disorder, characterized by recurrent attacks of a moderate to severe throbbing unilateral headache with symptoms such as nausea, vomiting, photophobia, osmophobia, and/or

Accepted for publication August 8, 2013.

*Conflict of Interest:* The authors declare that there is no conflict of interest.

phonophobia and in some cases preceded by neurological premonitory symptoms.<sup>1-3</sup> Indeed, in addition to complex neuronal spinal, supraspinal, and cortical mechanisms,<sup>1,7</sup> several endogenous and exogenous triggers, as well as genetic and epigenetic factors, are involved.<sup>2,3</sup> Taken collectively, the number of molecular and anatomical targets that we could manipulate in order to alleviate this disorder is broader.

In this context, although it has been claimed for a long time that the intervention of multiple systems simultaneously could be harmful, the design of specific drugs capable of activating several specific signaling pathways (multitarget drug approach) may avoid this problem. In short, this concept refers to the ability of a molecule (instead of a mixture of different molecules) to selectively target multiple receptors and/or mechanisms related to specific clinical conditions. Certainly, the multitarget drug approach or network pharmacology<sup>9,10</sup> could emerge as a new step in the development of innovative migraine pharmacotherapy.

Recently, the potential antimigraine compound, NXN-188, was designed with the objective of: (1) inhibiting the neuronal nitric oxide synthase and (2) activating the 5-HT<sub>1B/1D</sub> receptors,<sup>11,12</sup> both mechanisms strongly related to antimigraine activity.<sup>4</sup> Therefore, in addition to the current and future discovery of new molecules, anatomical structures, and pathways related to migraine pathophysiology, the design and development of a novel class of drugs capable of interacting with several (instead of a unique) targets, each of which are pivotal in this disorder, could help us to improve new therapeutic strategies.

Clinically, this idea is better illustrated with the use of the considered “dirty” or “promiscuous” drug, dihydroergotamine.<sup>4,5</sup> Admittedly, its use can be limited because of unwanted side effects, but in retrospect, the use of dihydroergotamine remains suitable as it is effective. Indeed, we could infer that this “old medicine” remains as an effective acute care medication because it acts via modulation of multiple family receptors (5-HT<sub>1</sub> receptors,  $\alpha_2$ -adrenoceptors, and D<sub>2</sub>-like receptors) rather than single targets associated with migraine pathophysiology.<sup>4,5,13</sup> We could propose that this heterogeneity can differentially

activate not only several receptors but also several specific signaling pathways (functional selectivity or biased signaling) with distinct efficacies and potencies<sup>14</sup> with critical therapeutic implications. This hypothesis is clearly depicted in the recent elegant studies of Wacker et al<sup>15</sup> and Wang et al,<sup>13</sup> where they demonstrated that the well-known unspecific 5-hydroxytryptamine receptor ligands, ergotamine (an antimigraine compound), serotonin (the endogenous ligand), and lysergic acid diethylamide (a psychedelic drug) are able to differentially (biased signaling) activate divergent signaling pathways in the same receptor.<sup>14,16</sup>

Thus, the design, discovery, and development of new drugs that reach several targets or specific signaling pathways involved in the migraine pathophysiology is essential to progress in the treatment of migraine and open a new field of study about the foremost pathways and targets that could synergistically improve the migraine management. This point of view could change the current paradigm of the “magic bullet” in the migraine treatment and point out the multitarget drug therapy as a new standpoint for encompassing the role of different systems involved in this complex neurovascular disorder. In this regard, the rational drug design of antimigraine molecules capable of interacting with several and specific targets remain as the new challenge to conquer.

*Acknowledgments:* AGH gratefully acknowledges the financial support of a Postdoctoral Fellowship awarded by the National Autonomous University of Mexico (DGAPA-UNAM).

## REFERENCES

1. Akerman S, Holland PR, Goadsby PJ. Diencephalic and brainstem mechanisms in migraine. *Nat Rev Neurosci.* 2011;12:570-584.
2. Vecchia D, Pietrobon D. Migraine: A disorder of brain excitatory-inhibitory balance? *Trends Neurosci.* 2012;35:507-520.
3. Levy D, Straaaman AM, Burstein R. A critical view on the role of migraine triggers in the genesis of migraine pain. *Headache.* 2009;49:953-957.
4. Mehrotra S, Gupta S, Chan KY, et al. Current and prospective pharmacological targets in relation to antimigraine action. *Naunyn Schmiedeberg Arch Pharmacol.* 2008;378:371-394.

5. Rapoport AM. What happens to the old headache medicines? *Headache*. 2012;52:701-706.
6. Viana M, Genazzani AA, Terrazzino S, Nappi G, Goadsby PJ. Triptan nonresponders: Do they exist and who are they? *Cephalalgia*. 2013;33:891-896.
7. Nosedá R, Burstein R. Migraine pathophysiology: Anatomy of the trigeminovascular pathway and associated neurological symptoms, CSD, sensitization and modulation of pain. *Pain*. 2013;S0304-3959. doi:10.1016/j.pain.2013.07.021.
8. Condés-Lara M, Rojas-Piloni G, Martínez-Lorenzana G, Díez-Martínez DC, Rodríguez-Jiménez J. Functional interactions between the paraventricular hypothalamic nucleus and raphe magnus. A comparative study of an integrated homeostatic analgesic mechanism. *Neuroscience*. 2012;209:196-207.
9. Csermely P, Agoston V, Pongor S. The efficiency of multi-target drugs: The network approach might help drug design. *Trends Pharmacol Sci*. 2005;26:178-182.
10. Hopkins AL. Network pharmacology: The next paradigm in drug discovery. *Nat Chem Biol*. 2008;4:682-690.
11. Vaughan D, Speed J, Medve R, Andrews JS. Safety and pharmacokinetics of NXN-188 after single and multiple doses in five phase I, randomized, double-blind, parallel studies in healthy adult volunteers. *Clin Ther*. 2010;32:146-160.
12. Bhatt DK, Gupta S, Jansen-Olesen I, Andrews JS, Olesen J. NXN-188, a selective nNOS inhibitor and a 5-HT<sub>1B/1D</sub> receptor agonist, inhibits CGRP release in preclinical models. *Cephalalgia*. 2013;33:87-100.
13. Wang C, Jiang Y, Ma J, et al. Structural basis for molecular recognition at serotonin receptors. *Science*. 2013;340:610-614.
14. Kenakin T. Functional selectivity and biased receptor signalling. *J Pharmacol Exp Ther*. 2011;336:296-302.
15. Wacker D, Wang C, Katritch V, et al. Structural features for functional selectivity at serotonin receptors. *Science*. 2013;340:615-619.
16. Reiter E, Ahn S, Shukla AK, Lefkowitz RJ. Molecular mechanism of  $\beta$ -arrestins-biased agonism at seven transmembrane receptor. *Annu Rev Pharmacol Toxicol*. 2012;52:179-197.